

PLINK GWAS - Data Quality Control: Questions

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Table 1: Sex check

FID	IID	PEDSEX	SNPSEX	STATUS	F
NA20506	NA20506				
NA20530	NA20530				
NA20766	NA20766				
NA20771	NA20771				
NA20757	NA20757				

Question 1: Why do you expect the homozygosity rate to be higher on the X chromosome in males than females?

Table 2: Duplicate and Related Individuals

FID1	IID1	FID2	IID2	Z(0)	Z(1)	Z(2)	PI_HAT
F1D1- Family ID for 1st individual; ID1 - Individual ID for 1st individual; F1D2- Family ID for 2nd individual; ID2 - Individual ID for 2nd individual; Z(0)- P(IBD=0); Z(1)- P(IBD=1); Z(2)- P(IBD=2); PI_HAT-P(IBD=2)+0.5*P(IBD=1) (proportion IBD)							

Question 2: How many duplicate pairs do your find (**hint: Pi-Hat = ~1**)? Do pairs with a **Pi-Hat = ~1** have to be duplicate samples? What is another explanation? What proportion would you expect a parent/ child to share IBD? Can you find any such relationship

Table 3: Hardy-Weinberg Equilibrium

Cases			Controls		
SNP	Pvalue	Population(s)	SNP	Population(s)	Pvalue

